

**ABERDARE TECHNICAL COLLEGE**  
**INFORMATION COMMUNICATION TECHNOLOGY MODULE 3**  
**UNIT: NETWORK DESIGN AND MANAGEMENT**  
**UNIT CODE: 0612 451 07A**  
**TIME: 4 HOURS**  
**TOTAL 50 MARKS**

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**Instructions to Candidates**

- ✓ *Read the whole paper before you begin.*
- ✓ *Attempt **all** tasks. Tasks A to F are practical; Task G is a short oral/written check.*
- ✓ *You may use a simulator (e.g. Cisco Packet Tracer) or physical equipment, as directed by your examiner.*
- ✓ *Diagrams must be neat, ruled and clearly labelled. Hand-drawn or software-drawn diagrams are both accepted.*
- ✓ *Call the examiner to witness each working stage **[DEMO]** before moving on.*
- ✓ *Mobile phones and notes are not allowed..*

## Scenario

A small company, **BrightPath Consultants**, has moved into a new office with three departments: **Human Resources (HR)**, **Accounts**, and **Reception**. Each department has **one computer**. The company has **one USB printer** that must be shared by all departments, and an **internet connection** from an ISP (supplied through a modem/router).

You have been hired as the network technician. You must plan, install, configure and test a network that meets the requirements below.

## Equipment Provided

| Item                                                                        | Quantity |
|-----------------------------------------------------------------------------|----------|
| Desktop computers (HR-PC, ACC-PC, REC-PC)                                   | 3        |
| USB printer (with USB cable)                                                | 1        |
| Switch (8 or 24 port)                                                       | 1        |
| Router / wireless router with internet access (ISP simulated)               | 1        |
| UTP Cat5e/Cat6 straight-through patch cables                                | 5        |
| RJ45 connectors, crimping tool, cable tester (cabling task), cable stripper | 1 set    |
| Diagram sheets                                                              | 1        |

## Network Requirements

### Functional Requirements

The installed network **must**:

| ID  | Requirement                                                                                                                                                                                             |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR1 | Interconnect the three departments (HR, Accounts and Reception) so that all computers can communicate with each other.                                                                                  |
| FR2 | Allow all three departments to use ONE shared USB printer. The printer is connected by USB to one computer (the print host, REC-PC) and shared over the network so that HR and Accounts can also print. |
| FR3 | Allow all three departments to share internet access through a single connection.                                                                                                                       |
| FR4 | Allow file sharing between departments, with a shared folder that all three computers can access.                                                                                                       |

|     |                                                                                                                      |
|-----|----------------------------------------------------------------------------------------------------------------------|
| FR5 | Give every device a unique IP address in the same private address range, with a default gateway for internet access. |
| FR6 | Allow Accounts to keep a folder that ONLY Accounts and HR can access (Reception must be denied access).              |
| FR7 | Allow users to verify connectivity using standard tools (ping, ipconfig/ifconfig, tracert).                          |

### Non-Functional Requirements

The network **should**:

| ID   | Quality                   | Requirement                                                                                                                           |
|------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| NFR1 | Reliability               | Use a wired star topology so the failure of one cable affects only one device.                                                        |
| NFR2 | Performance               | Use at least 100 Mbps links (Fast Ethernet or better) on all connections.                                                             |
| NFR3 | Security                  | Protect router/switch management with passwords, change default credentials, and (if wireless is used) secure it with WPA2 or better. |
| NFR4 | Scalability               | Use an addressing plan that leaves room for at least 20 more devices, and a switch with spare ports.                                  |
| NFR5 | Manageability / Usability | Use meaningful device names, label cables and ports, and keep documentation up to date.                                               |
| NFR6 | Maintainability           | Use a consistent, documented addressing scheme (static addresses for the router and the computers).                                   |
| NFR7 | Cost-effectiveness        | Use only the equipment provided; no unnecessary additional devices.                                                                   |

## Tasks

### Task A: Physical Network Diagram

Draw a **well-labelled physical diagram** of the network. Your diagram must show:

- All three computers (HR-PC, ACC-PC, REC-PC), the USB printer, the switch, the router and the internet/ISP cloud.
- The USB cable connecting the printer to its host computer (REC-PC).
- Cable type used on each link (e.g. Cat6 UTP straight-through).
- Switch port numbers and router interface names used for each connection.
- A title, a legend/key, your name, and the date.

### Task B: Logical Topology Design

1. Draw the **logical topology** showing how data flows between devices: IP addresses, the subnet, the default gateway, the path to the internet, and how HR-PC and ACC-PC reach the printer through the print host (REC-PC).
2. State the **physical topology** and the **logical topology** used (e.g. star physical, broadcast logical) and give **two reasons** for your choice.

### Task C: Addressing Table

Using the private network **192.168.10.0/24**, prepare an addressing table with these columns:

| Device | Interface / Port | IP Address | Subnet Mask | Default Gateway | Static / DHCP |
|--------|------------------|------------|-------------|-----------------|---------------|
|        |                  |            |             |                 |               |
|        |                  |            |             |                 |               |
|        |                  |            |             |                 |               |

Rules:

- Include the router (LAN and WAN), switch (management, if applicable), HR-PC, ACC-PC and REC-PC. (The USB printer has no network address; show it on the physical diagram only.)

- The router LAN interface must be the **first usable address**.
- The router must use a **static** address.
- Reserve a block for future growth and state it clearly.
- State the network address, broadcast address, number of usable hosts and CIDR notation.

#### Task D: Installation and Configuration

1. **Cabling:** Connect all devices according to your diagram. Label each cable and port. If crimping, make and test one straight-through cable and show the tester result.
2. **Router configuration:** Set the LAN IP, enable DHCP (if used), enable NAT/internet sharing, set the router hostname, and set a management password.
3. **Computer configuration:** Rename the computers (HR-PC, ACC-PC, REC-PC), assign IP settings per your table, and set the same workgroup (e.g. BRIGHTPATH).
4. **Printer sharing:** Connect the USB printer to REC-PC, install its driver, and share it. Add the shared printer on HR-PC and ACC-PC, then print a test page from each department.
5. **File sharing:** Create a shared folder named “CompanyShare” accessible to all three computers. Create a second folder, “FinanceHR”, accessible only to Accounts and HR users. Prove that Reception is denied access.
6. **Internet sharing:** Show that all three computers can browse the internet (or reach the simulated ISP server) through the shared connection.

#### Task E: Testing and Verification

Complete the test log below and demonstrate each test to the examiner.

| No. | Test                          | Action / Command | Expected Result  | Actual Result | Pass / Fail |
|-----|-------------------------------|------------------|------------------|---------------|-------------|
| 1   | HR-PC pings ACC-PC            | ping             | Replies received |               |             |
| 2   | ACC-PC pings REC-PC           | ping             | Replies received |               |             |
| 3   | REC-PC pings HR-PC            | ping             | Replies received |               |             |
| 4   | Each PC pings default gateway | ping             | Replies received |               |             |

|    |                                            |                               |                     |  |  |
|----|--------------------------------------------|-------------------------------|---------------------|--|--|
| 5  | Shared printer visible on HR-PC and ACC-PC | Add printer / browse \\REC-PC | Printer listed      |  |  |
| 6  | Each PC reaches internet/ISP               | ping / browse                 | Success             |  |  |
| 7  | Trace route to external address            | tracert                       | First hop = gateway |  |  |
| 8  | Print a test page from each PC             | Print                         | Page prints         |  |  |
| 9  | Access CompanyShare from each PC           | Open \\<IP>                   | Accessible          |  |  |
| 10 | Reception opens FinanceHR                  | Open \\<IP>                   | Access denied       |  |  |

Also record the output of **ipconfig /all** (or **ifconfig**) for **one** computer.

### Task F: Troubleshooting

The examiner will introduce **one fault** into your working network (for example: wrong subnet mask, disconnected cable, wrong gateway, or the print host being switched off). You must:

1. Identify the symptom and describe the troubleshooting steps you followed.
2. State the cause of the fault.
3. Fix the fault and re-test.

### Task G: Short Questions

1. What is the purpose of the default gateway?
2. Why must the computer connected to the USB printer stay switched on for other departments to print?
3. Name two ways to improve the security of this network.



## EXAMINER'S GUIDE

*Not for candidates – remove this section before issuing the paper*

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### Expected Physical Topology

ISP / Internet cloud → Router (WAN port) → Router (LAN port) → Switch → HR-PC, ACC-PC and REC-PC. The USB printer connects to REC-PC (print host) by USB cable and is shared over the network. This is a **star** topology with the switch at the centre. All network cables are straight-through UTP. Any computer may be accepted as print host if the candidate states and justifies it.

### Sample Addressing Table

**Network:** 192.168.10.0/24   **Mask:** 255.255.255.0   **Broadcast:** 192.168.10.255   **Usable hosts:** 254

| Device | Interface        | IP Address    | Subnet Mask   | Default Gateway | Assignment            |
|--------|------------------|---------------|---------------|-----------------|-----------------------|
| Router | LAN              | 192.168.10.1  | 255.255.255.0 | N/A             | Static                |
| Router | WAN              | From ISP      | From ISP      | From ISP        | ISP / DHCP            |
| Switch | VLAN 1<br>(mgmt) | 192.168.10.2  | 255.255.255.0 | 192.168.10.1    | Static                |
| HR-PC  | NIC              | 192.168.10.11 | 255.255.255.0 | 192.168.10.1    | Static / DHCP<br>res. |
| ACC-PC | NIC              | 192.168.10.12 | 255.255.255.0 | 192.168.10.1    | Static / DHCP<br>res. |
| REC-PC | NIC              | 192.168.10.13 | 255.255.255.0 | 192.168.10.1    | Static / DHCP<br>res. |

**Suggested reserved ranges:** .1–.9 network devices; .10–.49 static computers/servers (including future ones); .100–.200 DHCP pool for new devices. Any consistent, documented scheme is acceptable. DNS may be the router (192.168.10.1) or a public DNS such as 8.8.8.8.

### Marks Summary

| Task | Description                    | Marks |
|------|--------------------------------|-------|
| A    | Physical network diagram       | 15    |
| B    | Logical topology design        | 10    |
| C    | Addressing table               | 15    |
| D    | Installation and configuration | 30    |
| E    | Testing and verification       | 15    |
| F    | Troubleshooting                | 10    |
| G    | Short questions                | 5     |
|      | TOTAL                          | 100   |

### Marking Guide

| Task   | Mark allocation                                                                                                                                                                                                                                   |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A (15) | All devices shown (4); correct connections (4); cable types and port/interface labels (4); title, key and neatness (3).                                                                                                                           |
| B (10) | Logical diagram with IPs, subnet and gateway (5); topology named correctly (2); two valid reasons, e.g. easy fault isolation, easy expansion, centralised management (3).                                                                         |
| C (15) | All devices listed (4); valid, non-overlapping addresses in 192.168.10.0/24 (4); router uses first usable address and static assignments correct (3); network and broadcast address, CIDR and usable hosts stated (2); growth block reserved (2). |
| D (30) | 5 marks per sub-task (cabling, router, computers, printer, file sharing, internet). Give partial marks for partially working configurations. Deduct for default passwords left unchanged.                                                         |
| E (15) | Ten tests at 1 mark each for correct execution and recording (10); ipconfig /all output (2); accurate Pass/Fail judgement (3).                                                                                                                    |
| F (10) | Systematic diagnosis (4); correct cause (3); fix and re-test (3).                                                                                                                                                                                 |

### Task G Model Answers

1. It is the router address that devices send traffic to when the destination is outside their own network.
2. The USB printer has no network connection of its own; it is shared through that computer (the print host), so if it is off the printer is unreachable.
3. Change default passwords, use WPA2/WPA3, disable unused ports, enable the firewall, keep firmware updated (any two).
4. Use VLANs on a managed switch with inter-VLAN routing, or separate subnets on different router interfaces.
5. Network Address Translation lets many private addresses share one public IP address for internet access.

### **Suggested Fault Scenarios for Task F**

- Wrong subnet mask on one PC (reaches some devices but not others).
- Wrong default gateway (local network works, no internet).
- Print host (REC-PC) switched off, or printer sharing disabled (other PCs cannot print).
- Duplicate IP address between two computers (intermittent connectivity).
- Disconnected or faulty patch cable (no link light).
- Workgroup mismatch or disabled file and printer sharing (shared folder not visible).

### **Notes to Examiner**

- Prepare the simulated ISP/internet connection before the exam begins.
- Reset all equipment to factory defaults between candidates.
- Examiner sign-off is required at each [DEMO] stage. Record times on the candidate's score sheet.